

AMENDMENT NO. 5 JULY 1990
TO
IS 1824 : 1978 SPECIFICATION FOR INSECTICIDAL
SPACE SPRAY

(*Second Revision*)

(*Page 13, clauses B-1.1 to B-1.6*) — Substitute the following for the existing clauses:

'B-1.1 Apparatus

B-1.1.1 Spectrophotometer or Photoelectric Colorimeter — with a blue filter (420 nm) having 1-cm absorption cell.

B-1.2 Reagents

B-1.2.1 Malathion Reference Standard — of known purity.

B-1.2.2 Kerosene — distilled.

B-1.2.3 Acetonitrile — boiling range 80 to 82°C.

B-1.2.4 Ethanol — anhydrous.

B-1.2.5 Sodium Hydroxide Solution — 0.5 N.

B-1.2.6 Ferric Reagent

B-1.2.6.1 Stock solution — Dissolve 4.0 g of ferric chloride ($\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$) and 160 ml concentrated hydrochloric acid in water and dilute to 500 ml.

B-1.2.6.2 Dilute solution — Dilute 25 ml of the stock solution with water to 1 000 ml.

B-1.2.7 Carbon Tetrachloride — analytical reagent grade.

B-1.2.8 Copper Reagent — 1.5 percent (*m/v*). Prepare by dissolving 1.5 g of copper sulphate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) in 100 ml water.

B-1.3 Preparation of Standard and Sample Solutions

B-1.3.1 Preparation of Standard Solution

B-1.3.1.1 Weigh malathion reference standard of known purity and prepare accurately 1 percent (*m/v*) solution in kerosene. Take 5 ml of the solution with a pipette into a 100-ml volumetric flask, add 2.5 ml acetonitrile, shake for 5 seconds and make up the volume with ethanol.

B-1.4 Procedure

B-1.4.1 To obtain accurate results, it is important that the analysis of standard and sample solutions be carried out simultaneously through the following procedure without interruption.

B-1.4.2 Transfer a 5 ml aliquot each of malathion standard for analysis and sample for analysis into separate 250-ml separatory funnels. Add 2 ml sodium hydroxide solution and mix well by gentle swirling for 5-10 seconds. Allow the solution to stand for exactly 2 minutes. Immediately add 75 ml ferric reagent and mix well by swirling for 10 seconds. Allow the solution to stand exactly for 5 minutes. Add 50 ml carbon tetrachloride, allow the mixture to drain for 1 minute and add 2 ml copper reagent. Shake the mixture for exactly 1 minute.

B-1.4.3 Take clear solvent phase containing malathion standard and sample into two different absorption cells. Measure the absorbances at 420 nm using carbon tetrachloride as blank.

B-1.5 Calculation

$$\text{Malathion content, percent by mass} = \frac{A}{B} \times P$$

where

A = absorbance of sample for analysis solution at 420 nm;

B = absorbance of malathion reference standard solution for analysis at 420 nm; and

$$P = \frac{\text{Mass, in g, of malathion reference standard taken for preparation of 1 percent (m/v) solution.}}{\text{Mass, in g, of kerosene used in the preparation of 1 percent (m/v) solution of malathion reference standard}} \times \text{Percent purity of malathion reference standard'}$$

(FADC 1)